

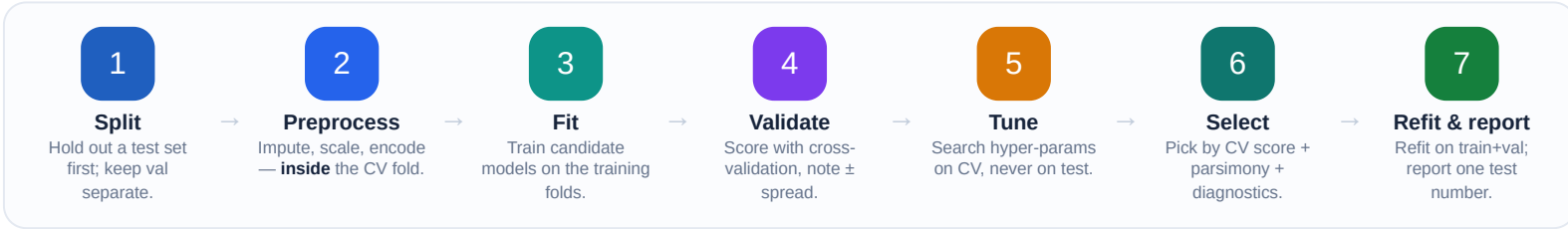
Build the Best Model: Workflow, Overfitting & Validation

The end-to-end path from raw data to a model you can trust — and the three numbers that tell you if it learned the signal or memorized the noise.

THE ONE RULE

The only score that estimates performance on **brand-new data** is **cross-validated**. Training and holdout are diagnostics — the CV mean $\pm$ spread is the truth.

THE END-TO-END MODELLING PIPELINE



🔒 Splitting & leakage guard

- ▶ **Train / Validation / Test** $\approx 60 / 20 / 20$. Touch test **once**, at the very end.
- ▶ **Stratify** on the label for classification so each split keeps class balance.
- ▶ **GroupKFold** when rows share a source (same slab/patient/user) — split by the group.
- ▶ **TimeSeriesSplit** for temporal data — train on the past, test on the future only.
- ▶ Fit scalers, imputers, encoders, and feature selection **inside** each fold (use a Pipeline).

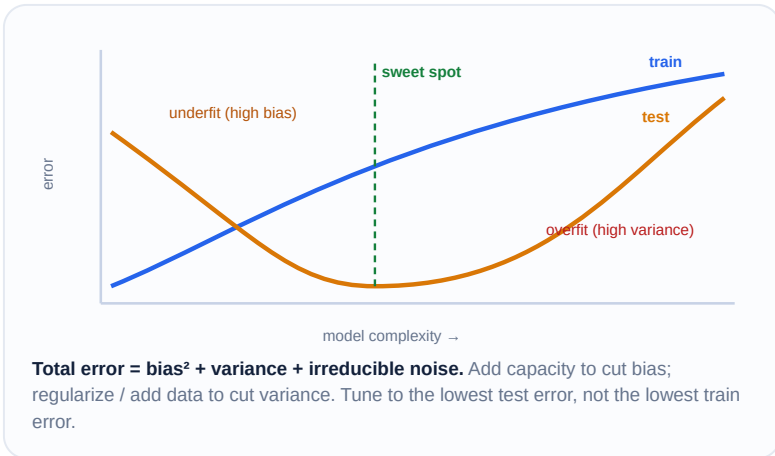
Leakage sources: target leakage (a feature encodes the answer), train $\rightarrow$ test contamination (preprocessing fit on all data), and temporal leakage (future info in past rows).

📏 The three R^2 — read them together

SCORE	WHAT IT IS	TRUST
Training R^2	Fit on data the model has already seen.	Diagnostic only
Holdout R^2	One untouched 20% split.	Quick check
Cross-val R^2	Averaged over k held-out folds, with $\pm$ spread.	The headline

Overfit: high train, big drop to CV, wide $\pm$. **Underfit:** low everywhere. **Healthy:** all close & high, tight $\pm$.

BIAS-VARIANCE & THE COMPLEXITY SWEET SPOT



Total error = bias² + variance + irreducible noise. Add capacity to cut bias; regularize / add data to cut variance. Tune to the lowest test error, not the lowest train error.

📈 Learning & validation curves

- ▶ **Learning curve** (score vs training-set size): gap that stays wide $\rightarrow$ variance, get more data / regularize. Both curves low & converged $\rightarrow$ bias, add capacity / features.
- ▶ **Validation curve** (score vs one hyper-parameter): find where CV peaks before it turns down.
- ▶ **Early stopping:** halt boosting / neural-net training when the validation score stops improving.

CROSS-VALIDATION SCHEMES — PICK THE ONE THAT MATCHES YOUR DATA

SCHEME	HOW IT WORKS	USE WHEN
k-Fold typically k = 5 or 10	Rotate each fold as the test set once; average the k scores.	Default for i.i.d. tabular data.
Stratified k-Fold	k-Fold that preserves class proportions in every fold.	Classification, especially imbalanced.
GroupKFold	No group appears in both train and test.	Repeated measures / clustered rows.
TimeSeriesSplit	Expanding window; always predict forward in time.	Any temporal / forecasting problem.
Repeated k-Fold	Run k-Fold several times with new shuffles; average.	Small data — tighten the estimate.
Nested CV outer + inner loops	Inner loop tunes hyper-params; outer loop measures the tuned model.	Honest score and tuning at once.

HYPER-PARAMETER SEARCH & CAPACITY CONTROL

🔧 Tuning strategies

- ▶ **Grid search** — exhaustive over a small, sensible grid.
- ▶ **Random search** — sample the space; wins when few params matter.
- ▶ **Bayesian optimization** (Optuna / Hyperopt) — model the score surface, spend trials where they help.
- ▶ **Successive halving** — start many configs cheaply, keep promoting the best.

Always tune against the **CV score**; keep the test set sealed.

🔴 Regularization (shrink to fight variance)

Ridge (L2): $\min \Sigma(y - \hat{y})^2 + \lambda \Sigma \beta_j^2$ — shrinks coefficients smoothly.

Lasso (L1): $\min \Sigma(y - \hat{y})^2 + \lambda \Sigma |\beta_j|$ — drives some β to 0 (feature selection).

Elastic Net: mixes L1 + L2 ($\lambda[\alpha \|\beta\|_1 + (1-\alpha)\|\beta\|^2]$).

Trees / NN: limit `max_depth`, `min_samples_leaf`, learning rate, subsample, dropout, and use early stopping.



TAKEAWAY

Split honestly, validate with cross-validation, tune on the CV score, then **refit and report one sealed test number**.
Fit and generalization are read together — never a single score.